

PHY517 / AST443: Observational Techniques

Homework 1

1. (1 pt) If you have not done so, fill out the pre-class survey.
2. (Not graded) Read the Course notes for PHY517 / AST443¹.
3. Read the wiki page on Grading² and the information in the links on plagiarism. Answer the following:
 - (a) (1 pt) How should you name the pdf file that you submit for this homework?
 - (b) (1 pt) You submitted your homework on time, but you notice an error before the deadline. What should you do?
 - i. Re-submit the entire homework with the error fixed, and delete the first submission.
 - ii. Submit a second pdf file with just the corrected part.
 - (c) (2 pts) Which of these are examples of plagiarism? (More than one answer may be correct.)
 - i. Copying your lab-mate's introduction section of a lab report.
 - ii. Taking somebody else's lab report, slightly modifying each sentence / paragraph, and submitting it as your own.
 - iii. Copying your buddy's telescope proposal for a Keck telescope, and submitting it as your own.
 - iv. Stating a "fact" from wikipedia without citing the original source.
 - v. Using ChatGPT to answer a homework question, following the links that ChatGPT provides to check for accuracy, and citing ChatGPT and the links as sources.
 - (d) (2 pts) What happens when you get a "Q" grade? (Answer yes/no/maybe.)
 - i. Immediate expulsion from school.
 - ii. You lose your scholarship.
 - iii. You receive an "F" for the course grade.
 - iv. You have to take the "Q" course.
 - (e) (1 pt) In this class, is the use of ChatGPT automatically considered plagiarism?
4. Read the wiki pages on Computing Resources³ and on Astro Software Overview⁴. Note that many of the tools mentioned on the latter are further described on individual wiki

¹https://github.com/sibirrer/PHY517_AST443/blob/main/documents/phy517_ast443_specifics.pdf

²https://github.com/sibirrer/PHY517_AST443/wiki/Grading

³https://github.com/sibirrer/PHY517_AST443/wiki/Computing-Resources

⁴https://github.com/sibirrer/PHY517_AST443/wiki/Astro-Software

pages on the course webpage, so be sure to check those for further clarification. In addition to reading these pages, answer the following questions:

- (a) (2 pts) Describe how you will log onto the Astro Computing Cluster. (Anticipate whether you expect to mainly log on from your own laptop, or from one of the machines in the lab.)
 - (b) (2 pts) Where should you put your data?
 - (c) (1 pt) Which software will we use to view images in fits format?
 - (d) (1 pt) Which software can you use to manipulate fits images?
 - (e) (1 pt) Which software can you use to figure out what area of the sky a given image covers?
 - (f) (1 pt) Where can you find a list of papers written about the Andromeda Galaxy?
5. (2 pts) As soon as your lab group has been assigned, **schedule the day-time part of the CCD Lab with the TAs**. This starts the clock for your Lab 1 report, so by scheduling it as soon as possible, you avoid your workload piling up later in the semester.
6. (2 pts) Your code returns a number of $99.123456789 \pm 0.00004556$ for your calculation. How should you report it in your lab write-up?
7. Let's practice finding an object:
- Convert a date of your choosing (e.g. your birthday) to a position on the sky using the following transformation:
 - Multiply the month birthday by 2. This number becomes the right ascension (if the result is 24^{h} , make it 0^{h}).
 - Subtract 2 from the day, and multiply the result by 3. This number becomes the declination.
 - Look up the resulting sky position on **simbad**⁵. Search for all objects within at least 0.5 degrees.
 - Sort the results by the number of references, and pick the most referenced object.
- (a) (2 pts) Make a finding chart for this object using the AAVSO finding chart tool⁶. The finding chart should be 15 degrees across. North should be up, and the chart should be orientated as if you were looking at the sky with the naked eye.
- (b) Use the ING StarAlt tool⁷ to determine when your object is best visible from Stony Brook. The higher up in the sky it is, the better visible it is. Note that Mt Stony Brook is not a predefined option in StarAlt, so you have to enter the coordinates manually. Pay attention to the format! Also note that the observatory coordinates have to be positive floats. For the following questions, make sure to include the relevant figure in your homework submission.

⁵<http://simbad.u-strasbg.fr/simbad/sim-fcoo>

⁶<https://www.aavso.org/apps/vsp/>

⁷<http://catserver.ing.iac.es/staralt/index.php>

- i. (2 pts) Use the ‘StarMult’ mode to determine the optimal observing date.
- ii. (2 pts) Use the ‘StarObs’ mode to see how the observability changes throughout the year. In which months is the target higher than 40° in the middle of the night?
- iii. (2 pts) Set the observing date to the optimal date, and make the ‘StarAlt’ plot. For how many hours is the target higher than 40° in this night? How far away from the Moon is it when it culminates? How much is the Moon illuminated?
- iv. (2 pts) Choose a date 6 weeks later, and make another ‘StarAlt’ plot. What changed? Are both nights good nights to observe your target, or is one clearly better?